

A Technical Introduction to Hidden Markov model

$$P(O|\lambda) = \sum_Q P(O|Q, \lambda) P(Q|\lambda)$$

Section 1

The choice of the urn for the n -th ball depends only upon a random number and the choice of the urn for the $(n - 1)$ -th ball. The choice of urn does not directly depend on the urns chosen before this single previous urn. Therefore, this is called a Markov process. It can be described by the upper part of Figure 1.

Section 2

Terminology The states of the process X_n (resp. X_t) are called hidden states, and $P(Y_n \in A | X_n = x_n)$ (resp. $P(Y_t \in B | X_t \in B_t)$) is called emission probability or output probability.

Section 3

HMMs can be applied in many fields where the goal is to recover a data sequence that is not immediately observable (but other data that depend on the sequence are). Applications include:

Section 4

Computational finance Single-molecule kinetic analysis Neuroscience Cryptanalysis Speech recognition, including Siri Speech synthesis Part-of-speech tagging Document separation in scanning solutions Machine translation Partial discharge Gene prediction Handwriting recognition Alignment of bio-sequences Time series analysis Activity recognition Protein folding Sequence classification Metamorphic virus detection Sequence motif discovery (DNA and proteins) DNA hybridization kinetics Chromatin state discovery Transportation forecasting Solar irradiance variability

Section 5

Other extensions Yet another variant is the factorial hidden Markov model, which allows for a single observation to be conditioned on the corresponding hidden variables of a set of K independent Markov chains, rather than a single Markov chain. It is equivalent to a single HMM, with N^K states (assuming there are N states for each chain), and therefore, learning in such a model is difficult: for a

sequence of length T , a straightforward Viterbi algorithm has complexity $O(N^2 K T)$. To find an exact solution, a junction tree algorithm could be used, but it results in an $O(N^{K+1} K T)$ complexity. In practice, approximate techniques, such as variational approaches, could be used. All of the above models can be extended to allow for more distant dependencies among hidden states, e.g. allowing for a given state to be dependent on the previous two or three states rather than a single previous state; i.e. the transition probabilities are extended to encompass sets of three or four adjacent states (or in general K adjacent states). The disadvantage of such models is that dynamic-programming algorithms for training them have an $O(N^K T)$ running time, for K adjacent states and T total observations (i.e. a length- T Markov chain). This extension has been widely used in bioinformatics, in the modeling of DNA sequences. Another recent extension is the triplet Markov model, in which an auxiliary underlying process is added to model some data specificities. Many variants of this model have been proposed. One should also mention the interesting link that has been established between the theory of evidence and the triplet Markov models and which allows to fuse data in Markovian context and to model nonstationary data. Alternative multi-stream data fusion strategies have also been proposed in recent literature, e.g., Finally, a different rationale towards addressing the problem of modeling nonstationary data by means of hidden Markov models was suggested in 2012. It consists in employing a small recurrent neural network (RNN), specifically a reservoir network, to capture the evolution of the temporal dynamics in the observed data. This information, encoded in the form of a high-dimensional vector, is used as a conditioning variable of the HMM state transition probabilities. Under such a setup, eventually is obtained a nonstationary HMM, the transition probabilities of which evolve over time in a manner that is inferred from the data, in contrast to some unrealistic ad-hoc model of temporal evolution. In 2023, two innovative algorithms were introduced for the Hidden Markov Model. These algorithms enable the computation of the posterior distribution of the HMM without the necessity of explicitly modeling the joint distribution, utilizing only the conditional distributions. Unlike traditional methods such as the Forward-Backward and Viterbi algorithms, which require knowledge of the joint law of the HMM and can be computationally intensive to learn, the Discriminative Forward-Backward and Discriminative Viterbi algorithms circumvent the need for the observation's law. This breakthrough allows the HMM to be applied as a discriminative model, offering a more efficient and versatile approach to leveraging Hidden Markov Models in various applications. The model suitable in the context of longitudinal data is named latent Markov model. The basic version of this model has been extended

to include individual covariates, random effects and to model more complex data structures such as multilevel data. A complete overview of the latent Markov models, with special attention to the model assumptions and to their practical use is provided in

Section 6

Concepts Teif, V. B.; Rippe, K. (2010). "Statistical-mechanical lattice models for protein-DNA binding in chromatin". *J. Phys.: Condens. Matter.* 22 (41) 414105. arXiv:1004.5514. Bibcode:2010JPCM...22O4105T. doi:10.1088/0953-8984/22/41/414105. PMID 21386588. S2CID 103345. A Revealing Introduction to Hidden Markov Models by Mark Stamp, San Jose State University. Fitting HMM's with expectation-maximization – complete derivation A step-by-step tutorial on HMMs Archived 2017-08-13 at the Wayback Machine (University of Leeds) Hidden Markov Models (an exposition using basic mathematics) Hidden Markov Models (by Narada Warakagoda) Hidden Markov Models: Fundamentals and Applications Part 1, Part 2 (by V. Petrushin) Lecture on a Spreadsheet by Jason Eisner, Video and interactive spreadsheet

Section 7

Probability of an observed sequence The task is to compute in a best way, given the parameters of the model, the probability of a particular output sequence. This requires summation over all possible state sequences: The probability of observing a sequence

Section 8

Filtering The task is to compute, given the model's parameters and a sequence of observations, the distribution over hidden states of the last latent variable at the end of the sequence, i.e. to compute $P(x(t) \mid y(1), \dots, y(t))$. This task is used when the sequence of latent variables is thought of as the underlying states that a process moves through at a sequence of points in time, with corresponding observations at each point. Then, it is natural to ask about the state of the process at the end. This problem can be handled efficiently using the forward algorithm. An example is when the algorithm is applied to a Hidden Markov Network to determine $P(h_t \mid v_{1:t})$.